

# Erdős Problem #40

EP-40

L1 - TRACTABLE

OPEN

Combinatorics

## ORIGINAL PROBLEM STATEMENT

For what functions  $g(N) \rightarrow \infty$  is it true that  $|A \cap \{1, \dots, N\}| \gg \frac{N^{1/2}}{g(N)}$  implies  $\limsup 1_A * 1_A(n) = \infty$ ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>